Sadman Rahman

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LinkedIn | Google Scholar | ResearchGate | Portfolio | GitHub

RESEARCH INTERESTS

- **Machine & Deep Learning:** Representation Learning, Sequence & Temporal Modeling, Multimodal Learning & Sensor Fusion, Lightweight / Efficient Deep Learning (Edge AI), Explainable AI.
- **Signal Processing:** Time–Frequency Analysis, Non-Stationary Signal Analysis, Wavelet & Multiresolution Analysis, Signal Decomposition, Adaptive Filtering.
- **Biomedical & Human-Centered AI:** Brain–Computer Interfaces, Human Activity & Gesture Recognition, Digital Health & Neurological Monitoring, Wearable Physiological Sensing.
- **Sensing Modalities:** Biosignals (EEG, ECG, sEMG, MMG), Inertial Sensing (IMU), FMCW Radar Micro-Doppler Sensing, Bio-Acoustic Signal Processing.

EDUCATION

Bangladesh University of Engineering and Technology (BUET) — Dhaka, Bangladesh
M.Sc. in Electrical and Electronic Engineering (EEE) — Oct 2026 – Ongoing

- **Supervisor:** Dr. Shaikh Anowarul Fattah.
- **Major:** Communication and Signal Processing
- **Relevant Courses:** Digital Image Processing, Quantum Computing, Machine Learning and Pattern Recognition

Military Institute of Science and Technology (MIST) — Dhaka, Bangladesh
B.Sc. in Electrical and Electronics Engineering (EEE) — Aug 2022 – Jun 2026

- CGPA **3.94/4.00**, ranked **2nd of 97**; MIST Dean's List and Merit Scholarship in all four years.
- **Supervisor:** Dr. Kazi Newaj Faisal.
- **Thesis:** Application-Aware Deep Learning Optimization for Processing 2D Joint-Variable Representations. **[Slides]**

Milestone College — Dhaka, Bangladesh
Higher Secondary Certificate (HSC), Science — Mar 2019 – Jan 2022

- Result: **GPA 5.00/5.00**.

PUBLICATIONS

14 publications to date: 6 journal articles (3 Q1, 2 Q2, 1 Q3) and 8 IEEE conference papers.

Journal Articles

[1] **Single-Channel Wearable EEG for ADL Recognition in Healthy and Epileptic Cohorts Using MODWT-MRA and Machine Learning.**
Mahdia Binte Maksud, **Sadman Rahman**, Md. Mehedi Hasan Bhuiyan, Tahiya Iffat Majumder, Kazi Newaj Faisal
Biomedical Signal Processing and Control ((Q1, IF: 5.7) · **[Accepted]** [pdf] 2026

[2] **A comparative logic-style framework for 8-Bit 12-function efficient ALU design using transmission gate and NMOS pass transistor logic in 90-nm CMOS.**
Saquib Farhan, **Sadman Rahman**, Sanjida Sultana, Md. Tawfiq Amin, Gourab Datta
Analog Integrated Circuits and Signal Processing (Q3, IF: 1.8) · **[Published]** [pdf] 2026

[3] **Radar-Based Sleep Posture Transition Sensing: An Efficient Deep Learning Hierarchical Temporal Architecture Utilizing Depthwise Convolution.**
Sadman Rahman, Rishi Raj Sharma, Kazi Newaj Faisal
IEEE Sensors Letters (Q2, IF: 2.4) · **[Published]** [pdf] 2026

[4] **Stacking Multi-Variate Time–Frequency Information: A Deep Learning Approach for IMU–MMG based Hand Gesture Recognition.**
Sadman Rahman, Md. Mehedi Hasan Bhuiyan, Nasiruddin Mahmud Himel, Rishi Raj Sharma, Kazi Newaj Faisal

IEEE Sensors Journal (Q1, IF: 4.5) · **[Published]** [pdf] 2026

[5] **HatSeqNet: A Sequence-based Deep Learning Model for Cross-Subject Driver Drowsiness Detection using Single-Channel EEG Signal.**
Sadman Rahman, Saquib Farhan, Rishi Raj Sharma, Kazi Newaj Faisal
IEEE Sensors Journal (Q1, IF: 4.5) · **[Published]** [pdf] 2026

[6] **Improved FMCW Radar Sensing for HAR Systems: Synchroextracting Transform-Enhanced Time–Frequency Representation and Custom Anisotropic CNN.**
Arnob Ghosh, **Sadman Rahman***, Labib Siddique Siam, Kazi Newaj Faisal
IEEE Sensors Letters (Q2, IF: 2.4) · **[Published]** [pdf] 2025

Selected Conference Proceedings

[7] **Short-term SVMD-Enhanced High-Resolution Time-Frequency Representation for ECG Signal Analysis.**
Sadman Rahman, Kazi Newaj Faisal
IEEE SPMB · **[Accepted]** 2026

[8] **FormSense: A Continuous Fatigue and Form Tracking Application for Resistance Workout Monitoring Utilizing Phase-Graded IMU Analysis.**
Abdullah Al Mahi, Munawar Arif Nitol, Md Mehedi Hasan Bhuiyan, **Sadman Rahman**, Kazi Newaj Faisal
IEEE ICACCESS · **[Accepted]** 2026

[9] **VoxScope: A Deployment-Oriented Analysis Framework for Extreme Vocal Distortion Techniques in Heavy Metal.**
Sadman Rahman, Kazi Newaj Faisal
IEEE SPICSCON · **[Accepted]** 2026

[10] **Exploring Gamma-Band Filtered Features and Machine Learning for Resource-Efficient Multi-Level Cognitive Load Classification Using EEG.**
Nur Md. Atiquzzaman Rasel, **Sadman Rahman***, Kazi Newaj Faisal
IEEE QPAIN **[Best Presenter Award]** · **[Published]** [pdf] 2026

[11] **Leveraging Power-Spectrum-based Features and Machine Learning for Resource-Efficient Cough Classification in Respiratory Disease Monitoring.**
Sadman Rahman, Irtiza Mohtasim, Saquib Farhan, Kazi Newaj Faisal
IEEE SPICSCON **[Best Paper Award]** · **[Published]** [pdf] 2025

[12] **Neural Network Enhanced PID Control of Automatic Voltage Regulators: A Comparative Study with Classical and Regression Approaches.**
Sadman Rahman, Saquib Farhan, Saiba Aabira, Sunjida Sultana
IEEE PEEIACON · **[Published]** [pdf] 2025

[13] **GCAF-Net: An Attention-Enhanced Deep CNN Framework with Grad-CAM for Classifying Rare and Endangered Medicinal Plants in Bangladesh.**
Sadman Rahman*, Saquib Farhan*, Saiba Aabira, Kazi Newaj Faisal
IEEE QPAIN · **[Published]** [pdf] 2025

[14] **High-Speed, Low-Power 8-Bit 12-Function ALU Using Transmission Gate Logic in 90nm Technology.**
Saquib Farhan*, **Sadman Rahman***, Saiba Aabira, Gourab Datta, Sunjida Sultana, Md. Tawfiq Amin
IEEE QPAIN · **[Published]** [pdf] 2025

*denotes equal contribution / co-first authorship

PROJECTS

- Developed an enhanced masked Wigner–Ville distribution for high-resolution time–frequency analysis of non-stationary biomedical signals. **[Submitted]**
- Developed a sharp, noise-robust time–frequency representation via synchroextracting Fourier masking. **[Submitted]**
- Developing D^2**-Net**, a radar–Doppler decomposition framework for human activity localization from FMCW micro-Doppler signatures. **[In preparation]**
- Supervised a smartphone-based induction-motor fault-diagnosis system using MEMS accelerometry and

Bayesian-optimized ML classifiers. **[Submitted]**

- Lead **JPR**, a localized, on-premise journal peer-reviewer assistant built on a Llama-3 RAG pipeline (LangChain, Ollama, ChromaDB). **[Ongoing]**
- Validated information-sharing network structures for joint modulation and SNR estimation via multi-task learning. **[Submitted]**
- Lead cluster-based feature integration for EEG motor-imagery classification using adaptive signal decomposition. **[Submitted]**
- Co-supervised activity-aware epilepsy discrimination from single-channel wearable EEG using MODWT-MRA and embedded ML. **[Submitted]**
- Co-lead an AI-enhanced human–machine interface for virtual reality using dual-hand IMU–sEMG sensor fusion. **[Submitted]**
- Developed an IMU-based attachable device for standard-pen signature authentication with embedded ML (MPU-6050). **[Submitted]**
- Developed lightweight convolutional patching networks for learning from 2D joint-variable representations. **[Submitted]**
- Lead Bangla abusive-speech detection for broadcast and communication systems using transformer-based audio classification. **[Completed]**

PROFESSIONAL EXPERIENCE

INSIA Lab — Dhaka, Bangladesh
Co-founder — May 2024 – Present

- Scale the team 16×, from 3 to 49 members, by establishing structured recruitment, onboarding, team allocation, and performance-review processes.
- Deliver 15 multidisciplinary projects end-to-end, by defining scope, allocating resources, tracking milestones, and coordinating technical teams.

Grameenphone (Telenor Group) — Dhaka, Bangladesh
Industrial Engineering Trainee — May 2025 – Jun 2025

- Studied operational workflows across the IT, Network Services, Central Operations, and Technology functions.
- Documented handoffs and escalation points in B2C service delivery.
- Wrote a KPI-based report and presented the findings to the supervising engineers.

ACADEMIC SERVICES

- **Course Instructor:** Taught Hands-on Arduino 3.0 and Pixel Pulse: A Rhythm of Creative Design, both run by the MIST Robotics Club and IEEE MIST SB, reaching 150+ students across the two courses.
- **Peer Reviewer:** IEEE Transactions on Affective Computing (Q1), reviewing submissions on affective computing and physiological signal analysis.
- **Teaching Interests:** Electrical Circuits, Electronic Circuits, Digital Logic Design, Signals and Systems, Digital Signal Processing, Microprocessors and Microcontrollers, and Machine Learning.
- **Conference Service:** Led student technical volunteers at IEEE RAAICON 2025; managed the 3-Minute Thesis and "Innovation for Smart Bangladesh" competitions at ICEEICT 2024.

LEADERSHIP AND ECA

IEEE MIST Student Branch — Dhaka, Bangladesh
Chairperson (Main Branch) — Jun 2025 – Jun 2026

- Launched 4 IEEE society chapters (SPS, EDS, MTT-S, PES); branch membership grew fivefold.
- Led 6 divisions and more than 100 volunteers, and ran 45+ technical events, 3 certification courses, and 2 international conferences.

IEEE Signal Processing Society, MIST SB Chapter — Dhaka, Bangladesh
Founding Chair and Vice Chair (Organizing) — Jun 2025 – Jun 2026

- Founded the chapter and organized 11+ technical workshops, working with BUET and DU.

MIST Robotics Club — Dhaka, Bangladesh
Secretary General — Jun 2025 – Jun 2026

- Secured sponsorship from Startech, Neural Semiconductor, and Walton for ROBOLUTION 2026, a national event with 1,000+ participants across 11 segments.
- Oversaw 30+ robotics events, including judging panels, equipment, and multi-track competitions.

Bangladesh National Semiconductor Symposium (BEAR) Dhaka, Bangladesh
BEAR Patron, MIST Representative Jun 2025 – Jul 2025

- Represented MIST at a 2,000+ attendee symposium and led a pavilion of 17 university research teams; the team placed 2nd Runner-up in the Biotech Poster Showcase.

AWARDS AND RECOGNITIONS

- **Best Paper Award**, IEEE SPICSCON 2025
- **Best Presenter Award**, IEEE QPAIN 2026
- **Finalist**, BEAR Summit & National Semiconductor Symposium, 2025
- **2nd Runner-up**, BEAR Summit Biotech Poster Showcase, 2025
- **2nd Runner-up**, IEEE Photonics & EDS Poster Designing Contest, 2024
- **1st Runner-up**, MIST Innovation Club "Inception to Graphic Design", 2023

TECHNICAL SKILLS

- **Programming:** Python, C, MATLAB, PyTorch, TensorFlow, Scikit-Learn, LangChain, LangGraph, RAG.
- **AI/ML:** Statistical Analysis, Deep Learning Architectures, Supervised Learning, Agent-based Modeling.
- **Electrical Tools:** Simulink, Proteus, Cadence, PSpice, Arduino.
- **AI agents:** LaTeX, Antigravity, Cursor, Git, Jupyter Notebook, Anaconda.
- **Design and Office:** Adobe Illustrator, Adobe Photoshop, Miro, MS Excel, Word, PowerPoint, Notion, Substack.

PROFESSIONAL REFERENCES

Reference 1: Dr. Kazi Newaj Faisal
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Reference 2: Dr. Md. Aminul Islam
Associate Professor, Dept. of EECE, MIST
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Reference 3: Dr. Md. Tawfiq Amin
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